



Continuum Mechanics

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Chapter 03

Continuum Mass and Force

Part 2



Recap: Traction, Stress, and Equilibrium

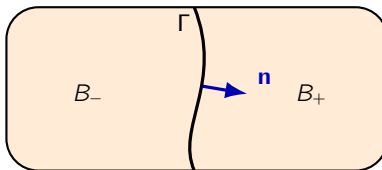
Traction Across an Internal Surface



Let an internal surface Γ divide the body locally into two disjoint open sides:

$$B \setminus \Gamma = B_- \cup B_+, \quad B_- \cap B_+ = \emptyset,$$

Choose the unit normal $\mathbf{n}$ to point from B_- toward B_+ .



The force per unit area exerted by B_+ on B_- is the traction

$$\mathbf{t}_n : \Gamma \longrightarrow \mathcal{V}.$$



Axiom 3.1 (Cauchy's Postulate)

There is a traction function

$$\mathbf{t} : \mathcal{N} \times B \longrightarrow \mathcal{V}$$

such that

$$\mathbf{t}_n(\mathbf{x}) = \mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x}).$$

Thus, at a point $\mathbf{x}$, the traction depends only on the point and the unit normal, not on the curvature of the surface through $\mathbf{x}$.

Action–Reaction



The principle of action–reaction requires

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = -\mathbf{t}(-\mathbf{n}, \mathbf{x}).$$

In words, the traction exerted by B_- on B_+ is equal and opposite to the traction exerted by B_+ on B_- .

This follows from the *Vanishing Boundary-Averaged Traction Condition*:

$$\frac{1}{\text{area}(\partial\Omega)} \int_{\partial\Omega} \mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x}) \, dA_{\mathbf{x}} \longrightarrow \mathbf{0} \quad \text{as } \text{vol}[\Omega] \longrightarrow 0. \quad (3.1)$$

We will return to this condition when studying the balance of linear momentum.



Result 3.3 (Cauchy's Theorem)

For each $\mathbf{x} \in B$, there is a second-order tensor $S(\mathbf{x}) \in \mathcal{V}^2$ such that

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = S(\mathbf{x})\mathbf{n}.$$

The field

$$S : B \longrightarrow \mathcal{V}^2$$

is called the *Cauchy stress field*.

Thus Cauchy's theorem replaces the normal-dependent traction function by a linear transformation acting on the normal.



Result 3.5 (Local Equilibrium Conditions)

At equilibrium, the absence of resultant force and resultant torque on every open subset of B is equivalent to

$$\begin{aligned}\nabla \cdot \mathbf{S}(\mathbf{x}) + \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) &= \mathbf{0}, & \mathbf{x} \in B, \\ \mathbf{S}^T(\mathbf{x}) &= \mathbf{S}(\mathbf{x}), & \mathbf{x} \in B.\end{aligned}\tag{3.8}$$

The first equation is the local force balance. The second states that the Cauchy stress tensor is symmetric at equilibrium.



Simple States of Stress



Definition (Spherical State of Stress)

Suppose that $S : B \rightarrow \mathcal{V}^2$ is the stress tensor for a body occupying $B \subseteq \mathbb{E}^3$.
If, at $\mathbf{x} \in B$,

$$S(\mathbf{x}) = -pI, \quad p \in \mathbb{R},$$

then a *spherical state of stress* exists at $\mathbf{x}$.

For every unit normal $\mathbf{n}$ through $\mathbf{x}$,

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = S(\mathbf{x})\mathbf{n} = -p\mathbf{n}.$$

Thus the traction is normal to every surface through $\mathbf{x}$ and has the same magnitude in every direction.



Definition (Uniaxial State of Stress)

At $\mathbf{x} \in B$, a state of stress is called *uniaxial* if there are a unit vector $\mathbf{d} \in \mathcal{N}$ and a scalar $\sigma \in \mathbb{R}$ such that

$$S(\mathbf{x}) = \sigma \mathbf{d} \otimes \mathbf{d}.$$

The state is called *pure tension* if $\sigma > 0$ and *pure compression* if $\sigma < 0$.

For a surface through $\mathbf{x}$ with unit normal $\mathbf{n}$,

$$\begin{aligned} \mathbf{t}(\mathbf{n}, \mathbf{x}) &= S(\mathbf{x})\mathbf{n} \\ &= (\mathbf{d} \cdot \mathbf{n})\sigma \mathbf{d}. \end{aligned}$$

Tractions in a Uniaxial State

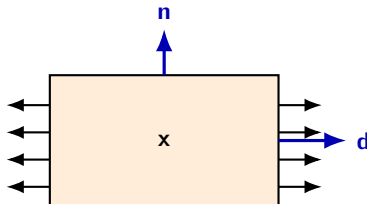


If $\mathbf{n} = \mathbf{d}$, then

$$\mathbf{t} = \sigma \mathbf{d}.$$

If $\mathbf{d} \perp \mathbf{n}$, then $\mathbf{d} \cdot \mathbf{n} = 0$, and hence

$$\mathbf{t} = \mathbf{0}.$$



The arrows show $\sigma > 0$. They reverse for $\sigma < 0$. Nonzero surface tractions are parallel to $\mathbf{d}$; tractions on surfaces whose normals are orthogonal to $\mathbf{d}$ vanish.



Definition (State of Pure Shear)

If, at $\mathbf{x} \in B$, there are orthogonal unit vectors $\mathbf{d}, \mathbf{m} \in \mathcal{N}$ and a scalar $\tau \in \mathbb{R}$ such that

$$\mathbf{d} \cdot \mathbf{m} = 0$$

and

$$S(\mathbf{x}) = \tau(\mathbf{d} \otimes \mathbf{m} + \mathbf{m} \otimes \mathbf{d}),$$

then a *state of pure shear* exists at $\mathbf{x}$.

For a surface through $\mathbf{x}$ with unit normal $\mathbf{n}$,

$$\begin{aligned} \mathbf{t} &= S\mathbf{n} \\ &= (\mathbf{m} \cdot \mathbf{n})\tau\mathbf{d} + (\mathbf{d} \cdot \mathbf{n})\tau\mathbf{m}. \end{aligned}$$



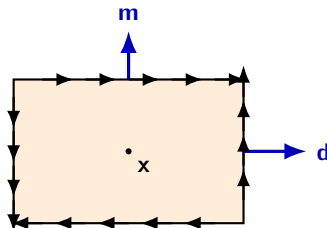
Tractions in a State of Pure Shear

If $\mathbf{n} = \mathbf{m}$, then

$$\mathbf{t} = \tau \mathbf{d}.$$

If $\mathbf{n} = \mathbf{d}$, then

$$\mathbf{t} = \tau \mathbf{m}.$$



The arrows show $\tau > 0$ and reverse for $\tau < 0$. The tractions are parallel to the boundary surfaces.



Example (Components of a Pure-Shear Stress)

Assume the right-handed orthonormal frame

$$\mathcal{F} = \{\mathbf{d}, \mathbf{m}, \mathbf{e}_3\}, \quad \mathbf{e}_3 = \mathbf{d} \times \mathbf{m},$$

and

$$\mathbf{S} = \tau(\mathbf{d} \otimes \mathbf{m} + \mathbf{m} \otimes \mathbf{d}).$$

Then

$$[\mathbf{S}]_{\mathcal{F}} = \begin{bmatrix} 0 & \tau & 0 \\ \tau & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$



Example (Components of a Pure-Shear Stress, Cont.)

For example,

$$\begin{aligned} S_{11} &= \mathbf{d} \cdot (S\mathbf{d}) \\ &= \mathbf{d} \cdot (\tau\mathbf{m}) = 0, \end{aligned}$$

$$\begin{aligned} S_{12} &= \mathbf{d} \cdot (S\mathbf{m}) \\ &= \mathbf{d} \cdot (\tau\mathbf{d}) = \tau, \end{aligned}$$

$$S_{13} = \mathbf{d} \cdot (S\mathbf{e}_3) = 0.$$

The remaining entries follow in the same way. Symmetry gives $S_{21} = S_{12} = \tau$, and the components involving $\mathbf{e}_3$ vanish.



Definition (State of Plane Stress)

If, at $\mathbf{x} \in B$, there are orthogonal unit vectors $\mathbf{d}, \mathbf{m} \in \mathcal{N}$ such that, in the right-handed orthonormal frame

$$\mathcal{F} = \{\mathbf{d}, \mathbf{m}, \mathbf{d} \times \mathbf{m}\},$$

the stress tensor has matrix

$$[S(\mathbf{x})]_{\mathcal{F}} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{21} & S_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

then a *state of plane stress* exists at $\mathbf{x}$.

Every state of pure shear is a state of plane stress.



Principal, Normal, and Shear Stresses



Definition (Principal Stresses and Principal Stress Directions)

Let $S : B \rightarrow \mathcal{V}^2$ be the stress tensor of a body in equilibrium. At $\mathbf{x} \in B$, the eigenvalues

$$\sigma_1(\mathbf{x}), \quad \sigma_2(\mathbf{x}), \quad \sigma_3(\mathbf{x}) \in \mathbb{R}$$

of $S(\mathbf{x})$ are called the *principal stresses*. The associated orthonormal eigenvectors

$$\mathbf{e}_1(\mathbf{x}), \quad \mathbf{e}_2(\mathbf{x}), \quad \mathbf{e}_3(\mathbf{x})$$

are called the *principal stress directions* at $\mathbf{x}$.

Thus,

$$S\mathbf{e}_i = \sigma_i\mathbf{e}_i, \quad i = 1, 2, 3,$$

with no summation on i .

Traction on a Principal Plane



For a surface through $\mathbf{x}$ whose normal is a principal direction $\mathbf{e}_i$,

$$\mathbf{t} = S\mathbf{e}_i = \sigma_i\mathbf{e}_i, \quad (\text{no sum on } i).$$

Hence the traction is normal to the surface.

Because the body is in equilibrium, $S = S^T$. Therefore the spectral theorem provides an orthonormal basis of principal stress directions, even when some principal stresses are repeated.



Definition (Normal and Shear Traction)

Let Γ be a surface through $\mathbf{x} \in B$ with unit normal $\mathbf{n}$, and let

$$\mathbf{t} = S\mathbf{n}$$

be its traction at $\mathbf{x}$. The *normal traction* is

$$\mathbf{t}_n := (\mathbf{t} \cdot \mathbf{n})\mathbf{n}.$$

The *shear traction* is

$$\mathbf{t}_s := \mathbf{t} - \mathbf{t}_n.$$

Thus,

$$\mathbf{t}_n + \mathbf{t}_s = \mathbf{t}, \quad \mathbf{t}_s \cdot \mathbf{n} = 0.$$



Definition (Normal and Shear Stresses)

The magnitudes

$$\sigma_n := |\mathbf{t}_n| \quad \text{and} \quad \sigma_s := |\mathbf{t}_s|$$

are called the *normal stress* and *shear stress*, respectively.



Example

Suppose that $\mathbf{n} = \mathbf{e}$ is a principal stress direction corresponding to the principal stress σ . Then

$$\mathbf{t} = S\mathbf{e} = \sigma\mathbf{e}.$$

Consequently,

$$\mathbf{t}_n = (\mathbf{t} \cdot \mathbf{e})\mathbf{e} = (\sigma\mathbf{e} \cdot \mathbf{e})\mathbf{e} = \sigma\mathbf{e} = \mathbf{t},$$

$$\mathbf{t}_s = \mathbf{0}.$$

Thus $\sigma_n = |\sigma|$ and $\sigma_s = 0$.



Maximal Normal and Shear Stresses



Result 3.6 (Maximal Normal and Shear Stresses)

Assume that, at $\mathbf{x} \in B$, the three principal stresses are distinct and ordered so that

$$\sigma_1 > \sigma_2 > \sigma_3. \quad (3.12)$$

Then the maximum value of the normal stress is

$$\max_{\mathbf{n} \in \mathcal{N}} \sigma_n = |\sigma_k| = \max\{|\sigma_1|, |\sigma_3|\},$$

where $k = 1$ or $k = 3$. This value is achieved for

$$\mathbf{n} = \pm \mathbf{e}_k.$$



Result 3.6 (Cont.)

Under the same assumptions, the maximum value of the shear stress is

$$\max_{\mathbf{n} \in \mathcal{N}} \sigma_s = \frac{1}{2} |\sigma_1 - \sigma_3|.$$

This value is achieved for the planes whose normals are

$$\mathbf{n} = \pm \frac{1}{\sqrt{2}} (\mathbf{e}_1 + \mathbf{e}_3) \quad \text{or} \quad \mathbf{n} = \pm \frac{1}{\sqrt{2}} (\mathbf{e}_1 - \mathbf{e}_3).$$



Proof.

Use the orthonormal principal frame and write an arbitrary unit normal as

$$\mathbf{n} = n_i \mathbf{e}_i, \quad n_1^2 + n_2^2 + n_3^2 = 1.$$

Since $S\mathbf{e}_i = \sigma_i \mathbf{e}_i$ (no sum on i), the traction is

$$\mathbf{t} = S\mathbf{n} = \sum_{i=1}^3 \sigma_i n_i \mathbf{e}_i.$$

Set

$$q := \mathbf{t} \cdot \mathbf{n} = \sum_{i=1}^3 \sigma_i n_i^2.$$

By definition,

$$\mathbf{t}_n = q\mathbf{n}, \quad \sigma_n = |q|.$$

Here i is a dummy index in each sum.



Proof (Cont.)

The coefficients n_i^2 are nonnegative and sum to 1. With $\sigma_1 > \sigma_2 > \sigma_3$, this gives

$$\begin{aligned} q - \sigma_3 &= (\sigma_1 - \sigma_3)n_1^2 + (\sigma_2 - \sigma_3)n_2^2 \\ &\geq 0, \end{aligned}$$

$$\begin{aligned} \sigma_1 - q &= (\sigma_1 - \sigma_2)n_2^2 + (\sigma_1 - \sigma_3)n_3^2 \\ &\geq 0. \end{aligned}$$

Hence $\sigma_3 \leq q \leq \sigma_1$ and

$$\sigma_n = |q| \leq \max\{|\sigma_1|, |\sigma_3|\}.$$

At $\mathbf{n} = \pm \mathbf{e}_1$ or $\mathbf{n} = \pm \mathbf{e}_3$, respectively, $q = \sigma_1$ or $q = \sigma_3$. This proves the normal-stress claim.



Proof (Cont.)

Since $\mathbf{t}_s = \mathbf{t} - q\mathbf{n}$ and $\mathbf{t}_s \perp \mathbf{n}$,

$$\begin{aligned}\sigma_s^2 &= |\mathbf{t}|^2 - |\mathbf{t}_n|^2 \\ &= \sum_{i=1}^3 \sigma_i^2 n_i^2 - q^2.\end{aligned}$$

For each $i = 1, 2, 3$, the ordering of the principal stresses implies

$$(\sigma_1 - \sigma_i)(\sigma_i - \sigma_3) \geq 0.$$

Expanding and rearranging,

$$\sigma_i^2 \leq (\sigma_1 + \sigma_3)\sigma_i - \sigma_1\sigma_3.$$

Multiply by n_i^2 and sum over i to obtain

$$\sum_{i=1}^3 \sigma_i^2 n_i^2 \leq (\sigma_1 + \sigma_3)q - \sigma_1\sigma_3.$$



Proof (Cont.)

Consequently,

$$\begin{aligned}
 \sigma_s^2 &\leq (\sigma_1 + \sigma_3)q - \sigma_1\sigma_3 - q^2 \\
 &= (\sigma_1 - q)(q - \sigma_3) \\
 &= \frac{(\sigma_1 - \sigma_3)^2}{4} - \left(q - \frac{\sigma_1 + \sigma_3}{2}\right)^2 \\
 &\leq \frac{(\sigma_1 - \sigma_3)^2}{4}.
 \end{aligned}$$

Thus

$$\sigma_s \leq \frac{1}{2}|\sigma_1 - \sigma_3|.$$

To finish, we identify the normals for which both inequalities are equalities.



Proof (Cont.)

Because $\sigma_1 > \sigma_2 > \sigma_3$, equality in

$$(\sigma_1 - \sigma_i)(\sigma_i - \sigma_3) \geq 0$$

holds for $i = 1, 3$ and is strict for $i = 2$. Therefore equality in the first shear-stress bound requires $n_2 = 0$.

Equality in the completed-square bound requires

$$q = \frac{\sigma_1 + \sigma_3}{2}.$$

With $n_2 = 0$, we have

$$n_1^2 + n_3^2 = 1, \quad q = \sigma_1 n_1^2 + \sigma_3 n_3^2.$$

Since $\sigma_1 \neq \sigma_3$, these equations give $n_1^2 = n_3^2 = \frac{1}{2}$. Thus the maximizing normals are

$$\mathbf{n} = \pm \frac{\mathbf{e}_1 + \mathbf{e}_3}{\sqrt{2}} \quad \text{or} \quad \mathbf{n} = \pm \frac{\mathbf{e}_1 - \mathbf{e}_3}{\sqrt{2}},$$

and $\sigma_s = \frac{1}{2}|\sigma_1 - \sigma_3|$ at each one.





Example (Pure Shear on a Rotated Plane)

Let $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be a right-handed orthonormal frame and consider the pure-shear stress

$$\mathbf{S} = \tau(\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1), \quad \tau > 0.$$

Choose a plane whose unit normal lies in the $\mathbf{e}_1$ – $\mathbf{e}_2$ plane:

$$\mathbf{n} = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2.$$

The traction on this plane is

$$\begin{aligned} \mathbf{t} = \mathbf{S}\mathbf{n} &= \tau[(\mathbf{e}_2 \cdot \mathbf{n})\mathbf{e}_1 + (\mathbf{e}_1 \cdot \mathbf{n})\mathbf{e}_2] \\ &= \tau(\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2). \end{aligned}$$



Example (Pure Shear on a Rotated Plane, Cont.)

The signed normal component of the traction is

$$\begin{aligned}\mathbf{t} \cdot \mathbf{n} &= \tau(\sin \theta \cos \theta + \cos \theta \sin \theta) \\ &= \tau \sin(2\theta).\end{aligned}$$

Thus

$$\mathbf{t}_n = \tau \sin(2\theta) \mathbf{n}, \quad \sigma_n = \tau |\sin(2\theta)|.$$

Set $\mathbf{m} = -\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2$, a unit tangent to the plane. Then

$$\begin{aligned}\mathbf{t} \cdot \mathbf{m} &= \tau(\cos^2 \theta - \sin^2 \theta) \\ &= \tau \cos(2\theta).\end{aligned}$$

Hence $\mathbf{t}_s = \tau \cos(2\theta) \mathbf{m}$ and $\sigma_s = \tau |\cos(2\theta)|$.



Example (Pure Shear on a Rotated Plane, Cont.)

At $\theta = 0$, the plane normal is $\mathbf{e}_1$. The traction is $\mathbf{t} = \tau \mathbf{e}_2$, so $\sigma_n = 0$ and $\sigma_s = \tau$.

At $\theta = \pi/4$,

$$\mathbf{n} = \frac{\mathbf{e}_1 + \mathbf{e}_2}{\sqrt{2}}, \quad \mathbf{t} = \tau \mathbf{n}.$$

The traction is entirely normal: $\sigma_n = \tau$ and $\sigma_s = 0$. Similarly, $\theta = -\pi/4$ gives the principal stress $-\tau$.

Also $S\mathbf{e}_3 = \mathbf{0}$, so the principal stresses are $\tau, 0, -\tau$. Result 3.6 therefore gives

$$\max \sigma_n = \tau, \quad \max \sigma_s = \frac{1}{2} |\tau - (-\tau)| = \tau,$$

exactly as the rotated-plane calculation shows. The maximum-shear planes at $\theta = 0$ and $\theta = \pi/2$ lie 45° from the principal directions.



Spherical and Deviatoric Stress Tensors



Proposition

Let $S : B \rightarrow \mathcal{V}^2$ be the stress tensor of a body, typically at equilibrium. Then there are second-order tensors $S_S, S_D \in \mathcal{V}^2$ such that

$$S = S_S + S_D,$$

$$\text{tr}(S_S) = \text{tr}(S), \quad \text{tr}(S_D) = 0,$$

and S_S is spherical. In particular,

$$S_S = -pl$$

for a scalar p .



Proof.

Set

$$p = -\frac{1}{3} \text{tr}(S) \quad \text{and} \quad S_S = -pl.$$

Then

$$\begin{aligned} \text{tr}(S_S) &= -p \text{tr}(I) \\ &= \frac{1}{3} \text{tr}(S) 3 \\ &= \text{tr}(S). \end{aligned}$$

Now set

$$S_D = S - S_S.$$

It follows that

$$\begin{aligned} \text{tr}(S_D) &= \text{tr}(S) - \text{tr}(S_S) \\ &= 0. \end{aligned}$$





Remark (Pressure and Principal Stresses)

At equilibrium, $S = S^T$, so its three real eigenvalues exist and

$$\text{tr}(S) = \sigma_1 + \sigma_2 + \sigma_3.$$

Hence

$$p = -\frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3).$$

The scalar p is called the pressure.

The sign convention makes $p > 0$ correspond to a compressive spherical stress state.



Definition (Spherical and Deviatoric Stress Tensors)

The tensor

$$S_S = -pI = \frac{1}{3} \text{tr}(S)I$$

is called the *spherical stress tensor*. The tensor

$$S_D = S - S_S = S - \frac{1}{3} \text{tr}(S)I$$

is called the *deviatoric stress tensor*.

Roughly speaking, the spherical stress changes the volume of a body but not its shape. The deviatoric stress changes the shape while producing little or no change in volume.



Definition (Deviatoric Stress Invariants)

Let S_D be the deviatoric part of an equilibrium stress tensor S . Define

$$J_2(S) := -I_2(S_D) = \frac{1}{2} S_D : S_D$$

and

$$J_3(S) := I_3(S_D).$$

Because $\text{tr}(S_D) = 0$, the second and third invariants of S_D describe the part of the stress state that is independent of the pressure.